

IB MATH AA SL

Lesson 37

Chain, Product & Quotient Rules

Calculus · 60-minute lesson

Syllabus SL 5.6 · Property of Lynda Badmus Education

d

Do Now — recall

ON YOUR WHITEBOARD

1. Differentiate x^3 .
2. Differentiate $(\text{stuff})^2$ — what's the outer power?
3. Differentiate $5x^2$.

Answers (click to reveal)

1. $3x^2$
2. $2(\text{stuff})$
3. $10x$

What we're learning today

LEARNING OBJECTIVES

- 1** **Apply**
the chain rule for composite functions.
- 2** **Apply**
the product rule.
- 3** **Apply**
the quotient rule.
- 4** **Choose**
the correct rule for an expression.

SUCCESS CRITERIA — I CAN...

- ✓
differentiate $(\dots)^n$ with the chain rule
- ✓
differentiate a product $u \cdot v$
- ✓
differentiate a quotient u/v
- ✓
decide which rule a function needs

The chain rule

COMPOSITE FUNCTIONS

$$\frac{d y}{d x} = \frac{d y}{d u} \cdot \frac{d u}{d x}$$

Outer × inner

differentiate the outside, then × derivative of the inside

Spot it

a 'function inside a function'

EXAMPLE

$$y = (2x+1)^5$$

$$\text{let } u = 2x+1$$

$$\frac{dy}{du}$$

$$= 5u^4$$

$$\frac{du}{dx}$$

$$= 2 \rightarrow \frac{dy}{dx} = 10(2x+1)^4$$

Chain rule

QUESTION

Differentiate $y = (3x + 2)^4$.

SOLUTION

Let $u = 3x + 2$, so $y = u^4$:

$$\frac{dy}{du} = 4u^3, \quad \frac{du}{dx} = 3$$

Multiply (chain rule):

$$\frac{dy}{dx} = 4(3x+2)^3 \times 3 = 12(3x+2)^3$$

Your turn — chain rule

ANSWER IN YOUR BOOK

1. $y = (x + 1)^3$. Find dy/dx .

2. $y = (2x)^5$. Find dy/dx .

3. $y = (5x - 1)^2$. Find dy/dx .

Answers (click to reveal)

1. $3(x+1)^2$

2. $10(2x)^4 = 160x^4$

3. $2(5x-1) \cdot 5 = 10(5x-1)$

Product & quotient rules

TWO MORE RULES

$$(u v)' = u' v + u v'$$

$$\left(\frac{u}{v}\right)' = \frac{u' v - u v'}{v^2}$$

Product

differentiate one, keep the other, and add

Quotient

'low d-high minus high d-low, over low squared'

WHICH RULE?

Inside a power
chain rule

Multiplied
product rule

Divided
quotient rule

Product rule

QUESTION

Differentiate $y = x^2(x + 1)$ using the product rule.

SOLUTION

$$u = x^2, v = x + 1 \rightarrow u' = 2x, v' = 1:$$

$$y' = u'v + uv' = 2x(x + 1) + x^2(1)$$

Simplify:

$$y' = 2x^2 + 2x + x^2 = 3x^2 + 2x$$

Exam-style question

Differentiate, simplifying where possible:

(a) $y = (4x - 3)^5$. [2]

(b) $y = x \cdot (2x + 1)$. [3]

MARK SCHEME — reveal after students attempt (tutor only)

(a) $\frac{dy}{dx} = 5(4x-3)^4 \times 4 = 20(4x-3)^4$ (M1)(A1)

(b) product: $1 \cdot (2x+1) + x \cdot 2 = 4x + 1$ (M1)(A1)(A1)

Common mistakes



Chain rule

Don't forget to $\times$ the derivative of the inside.



Product rule

$u'v + uv'$ — two terms, added.



Quotient order

Numerator is $u'v - uv'$ (mind the order/sign).



v^2

The quotient rule divides by v squared.

Mixed practice

QUESTIONS

1. $y = (x+5)^4$. dy/dx ?
2. $y = (3x-2)^3$. dy/dx ?
3. $y = x^2(x-1)$. dy/dx (product)?
4. Which rule for $(x+1)/(x-1)$?

Answers (click to reveal)

1. $4(x+5)^3$
2. $9(3x-2)^2$
3. $2x(x-1)+x^2 = 3x^2-2x$
4. quotient rule

Plenary & exit ticket

KEY FORMULAS

$$\frac{d y}{d x} = \frac{d y}{d u} \cdot \frac{d u}{d x}$$

$$(u v)' = u' v + u v'$$

EXIT TICKET — before you leave

1. $y = (x+2)^3$. dy/dx ?

2. Product rule formula?

3. Rule for $(...)^6$?

Answers (click to reveal)

1. $3(x+2)^2$

2. $u'v + uv'$

3. chain rule

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Chain, product & quotient rules.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 38

Stationary points & curve sketching.

$$f'(x) = 0$$